

environment and save precious water resources. Waterless solar panel cleaning is a new technology, which uses a dry cleaning technique to enhance the capability of solar energy harnessing system.

Unicorn is a smart robot with machine learning (ML) capabilities. It senses dust levels and cleans accordingly.

It can also differentiate dust from bird droppings or panel breakage. It features a rotary brush coupled with blower and auxiliary brushes to clean the panels. Unicorn can be set to operate at a fixed time of the day and week by using 'time' mode, or it can be operated based on the dust level when set in 'auto' mode. The fully automated system is ideally suited for both rooftops and ground mount farms, irrespective of the size and geography.

It sends a communication to the central monitoring station or designated mobile phone over GPRS when manual intervention is required. This intelligent robot can easily clean 400-800 panels per hour, depending upon the level of dust.

Shreem is exclusively designed for rooftop solar projects. It consists of two sets of soft microfibre cloth brushes (axial and radial). Controlled airflow ensures dust is blown away



from the panels efficiently. The radial brush also ensures it blows away any leftover dust from edges.

The robot weighs approximately 5kg, which makes it light and portable. It can clean around 200 panels an hour and comes with a backpack so that it can be carried from one location to the other. In its basic version, the Shreem robot is Bluetooth enabled and can be operated using a mobile app. In case of large farms, it can be operated using an RF remote-operated controller.

Both robots use unique airwash technology to ensure there is no need for water or any harmful chemicals. The docking station monitors relevant weather parameters like rain, wind speed, etc. If the parameters are unsafe for the operation of robots, these continue to be in the docking station. The robots are powered by Li-ion batteries.

The docking station is equipped

with a solar panel to charge the robot batteries when docked. There is no need for an external power supply or any cabling. The use of such technologies extends the life span of the panels, providing a payback of six to eighteen months, offering greater throughput, and increased efficiency.

Founded in 2017, the company is led by Suraj Vernekar, an engineer turned into an avid entrepreneur. "By 2025, an estimated 1.8 billion people will live in areas plagued by water scarcity, with two-thirds of the world's population living in water-stressed regions," says Suraj.

Aegeus is the only Indian startup that has received a patent for the innovative and unique airwash system. These waterless solar panel cleaning systems have received a nod from NCPRE-IIT Mumbai. The company strives to develop products that help clients to consume fewer resources.

The startup is also working on the Internet of Things (IoT) products like distribution transformer monitoring solution to monitor transformers, ensure continuity of power supply, and protect valuable assets of the utilities.

—Pallavi Singh

4 A CONNECTED SPORTS BIKE REDEFINING THE INDIAN EV ECOSYSTEM

With awareness regarding the benefits of electric mobility in maintaining a clean and green environment, it has gained drastic importance in India over the last few years. Now with the goal of self-reliance for reviving the country's economy, fostering Made in India technologies and replacing conventional fuel become far more crucial.

So a Bengaluru-based startup established by Niraj Rajmohan and Narayan Subramaniam in 2015, Ultraviolet Automotive unveiled F77, a

high-performance premium electric sports bike last year with an aim to boost the electric vehicle (EV) sector and bust the myths around it.

The company spent the initial years in building a business model and testing prototypes of motorcycles, chargers, and modular batteries through self-funded capital, starting with the first prototype completion in June 2016. It won the excellent design award at the International Bike Festival 2016 in Taipei for developing a hybrid cycle. In 2017, automobile com-

pany TVS bought 15 per cent stake in the startup, which grew to about 25 per cent later on.

The two-wheeler having a lightweight steel-aluminium frame uses a sealed motor and electronic modules. It has only hand controls and there is no clutch. The range varies according to the load, driving pattern, road conditions, ambient conditions such as temperature, and so on. Previous data from riders and vehicles over a period of time is fed into machine learning models, and range estimates